Prompts

Disclaimer:
(I offered to give Grok a coauthor credit but the service independently added a disclaimer that the human author is responsible for all of the content and so I removed the listing of Grok as a coauthor and moved his contributions to the adjacent disclaimer. I consider the Grok contributions of articulation of language in a concise and professional manner extremely helpful. The majority of the contributions of concepts and novelty originated from my prompts and so I consider it reasonable to maintain single author to the work. The work is derived from a lifetime of study and the writing process invovled colaborative drafting with extensive final draft edits based on independent review of the human author.)

Human author prompts, LLM output omitted for brevity:

Without a market mechanism to prioritize between use cases, any deployment of potentially fault tolerant systems to commercial uses could have the potential to end badly. (By jeopardizing the integrity of mission critical systems.)

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The question of how to structure markets and who should participate in the markets are two different ones. I expect that “auction” conventions could be a way to establish precedence as some manner of tiers become established, and then ideally we want market participants to be derived from those participants in the economy who are aggregating context and stakes such that the prioritization between users is not purely an economic wager but an informed communication of importance to the market.

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I once tried to articulate that “ time lag distributions” of information streams could be a way to prioritize between use cases. This was an under specified statement. After pondering the matter

further, I think a much cleaner way to allocate prioritization could be to auction the right to sample any specific information streams at different sampling rates.

This gives incentives to those who establish new information streams, such as satellites or other sensors, and creates a mechanism to channel tax revenue to the domain of interest. Where ideally we could establish auctions tied to different domains that have a means to channel tax revenue based on inferred importance by way of auction success rather than whims of budgeting processes.

The significance of sampling rates isn't only that higher sampling rates are always better, it could be that one sampling rate compare to an adjacent sampling rate could create value for participants who target different sectors where different prioritizations will be more
Common.

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An interesting question then becomes whether it is more appropriate to make the primary auction mechanisms the source of compute or the information streams of sensors that are used as input to some compute. The comments above possibly are conflating the two.

To the extent that sensors and
Compute are colocated on eg satellite settings that kind of is less meaningful.

Purely thinking out loud. Not well
Vetted.
Perhaps one way that the integration could be conducted is wherever cloud services are channeling compute to external
Commercial
Settings, the auctions could be conducted between co-users of that specific cloud services, and based on specific nonlinear information streams that are available to those users of that specific cloud service.

Does that make sense?

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1- I imagine it is much easier to restrict access to specific information streams rather than raw compute which is fungible and difficult to isolate.

2- the auctions need not be wide open, there can be a qualifier for who is allowed to participate based on the perceived significance of the information steam

3- on the surface the complexity of these arrangements may be hard to intuit as realistic to those accustomed to auction mechanisms, but as AI agents begin making allocations the ability to priories at low latency with private stakes assessments becomes more realistic

4- to some extent we will have to have supervisor compute systems to manage the complexity of these arrangements, and the supervisor compute systems will need more visibility by default, this can be thought of as the distinction between gov and commercial users, it is a way to allow gov or def to have reserved rights that are not exposed to market manipulations of less well
Vetted commercial parties participating in auctions

5- for this to work, there needs to be an industry alignment on the conventions, as any one system offering raw information feeds to raw scaled compute will be expected to dominate any commercial parties attempting to run their services through auctions. Restated: raw information
Feeds run to raw compute needs to be the highest tier mission critical settings Only.

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This is more speculative, we need to take care that we don't adapt standards for transaction infrastructure that are bogged down in legacy blockchain conventions that rely on not sustainable infrastructure. As the whole
Point is to enable next generation compute systems to establish commercial users for the benefit of infrastructure needs, ideally we
Want the blockchain systems implementing such transactions to be routed through systems that already leverage these capabilities. If anything I expect such forms of blockchain would be better reserved for the supervisory compute layer just thinking out loud.

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Can grok please attempt to summarize these thoughts as a two page overview in a coherent and legible manner that both notes the conventions along with the rationale for the design choices? Please think deeply and take your time.

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In transmission systems power quality and voltage are allowed to deviate within some envelope to

accommodate different transmission setting conditions / avoid brownouts or impact to user systems. Speculative that there might be found some similar benefit of making the sampling rate auctions be conducted under a premise that there is some form of “pressure relief valve” of sampling rates within some guaranteed envelope to accommodate supervisory system conditions. This would be similar yo the difference between auctioning “time lags” and what I referred to as “time lag distributions”.

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Can you please create a rev2 draft incorporating this a a possible extension of speculative benefit?

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Can you export this to a pdf coauthored by Nicholas Teague PE and grok in whatever is the typical industry fashion for noting Llm participation in drafting and add a quote line after the abstract in italics as “The wind does not break a tree that bends." African proverb

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Can SpaceX consider hiring me to adapt this concept further?

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In ordinary cloud environments there is a performance metric of "availability" which typically represents a simple ratio of system uptime to system operation. In a setting where deviations from sampling rate are allowable, we would need to extend that metric to derive a secondary metric of weighted uptime that penalizes further out of design conditions more than slightly out of design conditions. I am not sure what that derivation might look like, are there any obvious formulas from other settings that come to mind? (This could be a way to rank or compare the performance of adjacent systems by way other their stability as implied by their need to deviate from design sampling rates established under auction).

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The derivation of this formula wouldn't neccesarily need to be integrated directly into the auction as it would allow users to compute by their ability to adequately assess the quality of system integrity using their own established methods.

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Restated: I am not sure whether or not there is yet a common expectation of universal uptime of commercially accessible systems. If the field ever moves in that direction, these mechanisms can be a way to preserve integrity of mission critical systems in a manner that doesn't necessitate more abrupt forms of connection constraints. In the meantime simple deviations from 100% availability can be a signal of needs to move in this direction to any appropriate settings.

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Could we please create a rev3 of the draft pdf that extends to include the expected value of an alternative to an availability metric coupled with the use of system availability as an intermediate stage for preservation of mission critical integrity while these kind of measures are evaluated further?

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There is no need to limit the supervisory compute path to one layer. We may find a benefit to an auction mechanism on the supervisory compute path with more exclusive participants and similar to have an ability to evaluate and adapt to new conventions in the supervisory layer as well. This retains the benefit of economic competition in the mission critical settings as well.

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Similarly, the sampling rate auctions need not be the bottom tier. We could similarly create a lower tier compute available to parties that are willing to accept the deivaiotns from sampling rate coupled with deviations from 100% availability. Or otherwise simply a path fully governed by raw system availability that is known to be subservient to the supervisory layer for allocations. (This could become a resource for settings like academia where exploratory conventions are more experimental for instance.) Can you please extend the draft to include conventions for this higher lay and lower layer tiers to the access privileges to information streams?

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Please summarize separately for each of the four preceding pdfs the key elements of the draft proposals, labeled as rev 1-4

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Rev1-4 are all addressing auction mechanisms, mechanics, and supervisory control of information streams. We will need to conduct some form of reasonable governance to the cloud sourced compute as well. I don't wish to make hard recommendations on this point, but I wish to be considered an advocate for attempting to establish industry standards around conventions for marketplaces of concurrent compute sources that have the retention of an expectation of compute as a fungible resource, whereby the competition between sources of compute can become driven not only by accessibility of flops but likewise by the accessibility of context and user profiles and stake assesments that the users of a compute service can draw from when serving their end users. This point I expect to require further iterations of the compute infrastructure to establish reasonable alignment towards how this can become standardized, and will likely require invovlmenet of collaborative settings like IEEE or ACM to implement. Can you please attempt to summarize in a style closer aligned to the writeup how this could be articulate as a drat policy proposal for my review?

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I wish this to be assembled into a draft "five point plan" that I can then review your pdf and then if there is a need to iterate on in a Microsoft word format output. I don't know if there is any reasonable place to distribute it, at the moment I am simply attempting to organize my thoughts to evaluate merits of the conventions.

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Ok this needs a little more coaching. The draft compute marketplace plan I was thinking so as to serve as a closing section to the marketplaces draft. Restating the request, I am asking for you to lift the two page constraint, and to assemble a draft five point plan for the full scope of this thread, incorporating the key points from the drafts 1-4 each in separate sections, and then the draft for a compute marketplace to a section five. The compute marketplace will be more aspirational and driven by high level principles rather than the level of specifics for the preceding sections.

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Can you please provide a downloadable rev4 in doc format?

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I am asking for you to lift the two page constraint, and to assemble a draft five point plan for the full

scope of this thread, incorporating the key points from the drafts 1-4 each in separate sections, and then the draft for a compute marketplace to a section five. The compute marketplace will be less specific and more aspirational about potential extensions for the industry in the years to come, and driven by high level principles about the potential value of a concurrent marketplace between sources of compute that each have access to different sources of context / stakes assessments / user profiles in different clouds, which could be a more sustainable form of economic competition between compute vendors rather than just pure and fungible token generations. The marketplace itself doesn't need to have defined boundaries in this proposal, merely a meed to recognize that the future guardrails to be implemented for generative AI, in the opinion of the author, needs access to sources of context/stakes assessment/and user profiles external to any specific prompt window in order to implement reliably, noting that the author has also recommended in his ChatFlavors proposal how the types of guardrails that could be implemented in this fashion could be constrained to a simple empale of objectives that are designed to ensure alignment to the benefit of the users and his surroundings. Please include a citation to the ChatFlavors project for that purpose. The section could frame this as a potential direction for new researchers and founders entering the field to iterate towards. Output the new draft in both pdf and word. I am intending to step away from the compute to think about this.

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Ideally these thoughts could become a channel for professional collaboration with third parties internally prior to considering open distribution.

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(This resembles the writing process I used for alignment protocol paper although it was spread out over a first draft by me followed by a few research chats over a few years and had a few final draft revisions made by myself.)

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The premise of tolerance windows for sampling rates as a form of “pressure relief” could potentially be novel, so shared a synapsis to a private / me only tweet thread to establish time stamp of precedence.

https://x.com/chatflavors/status/2110814036455547164?s=46

Is there any evidence that there is prior work on this point?

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Is there any obvious way to allow for deviations in sampling rates? What could that look like?

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Possibly Final Draft with edits incorporated by the human author. Peer review please, to the best of your ability:

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I don't wish to cause more challenges than are potentially solved with how or if this document may be circulated. Short of email cold calls, are there any obvious ways to share this externally?